

Czech Technical University in Prague
FJFI - Department of Physics
02ELMA - Electricity and Magnetism
Midterm 1

11.04.2025 / 14:30pm

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 90 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. Use of any class materials and calculators are not allowed.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. Write your answers in the provided space. If you need more space, additional sheets will be provided.

Question 1

A solid insulating sphere of radius R carries a spherically symmetric, non-uniform charge distribution given by

$$\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right), \quad 0 < r < R,$$

and $\rho(r) = 0$ for $r \geq R$. Here, ρ_0 is a constant.

- (a) Calculate the electric field $\vec{E}(r)$ as a function of the distance r from the center of the sphere, for $0 < r < R$ and $r \geq R$. — *0.9 pts*
- (b) Do the expressions for $\vec{E}(r)$ you derived for inside and outside the sphere agree at $r = R$? — *0.1 pts*

Solution:

Question 2

Two large, finite, parallel conducting plates of area A are placed a distance d apart in vacuum. One plate carries a total charge $+Q$, and the other carries $-Q$. The plates are electrically isolated (i.e., no battery is connected) and no edge effects are considered (i.e., the field is always uniform between the plates). The electric field outside the region between the plates is negligible.

- (a) Find the electrostatic energy stored in the system by using the integral

$$U = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau,$$

which is the electrostatic energy stored in an electric field $\vec{E}$, where ϵ_0 is the permittivity of free space and $d\tau$ is a volume element. — *0.7 pts*

- (b) Suppose the distance between the plates is increased by a small amount Δx with the charges remaining constant. How much work must be done in this process? — *0.3 pts*

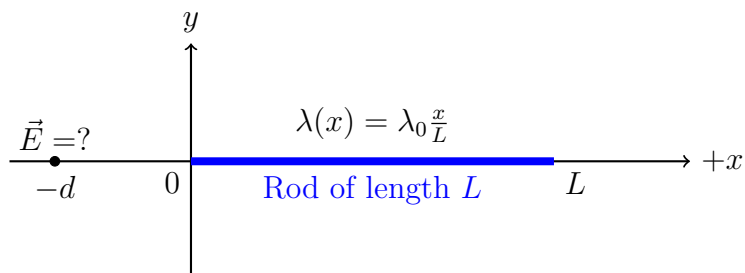
Solution:

Question 3

A thin rod of length L lies along the x -axis extending from $x = 0$ to $x = L$. The rod carries a nonuniform linear charge density given by

$$\lambda(x) = \lambda_0 \frac{x}{L}$$

where λ_0 is a constant with units of charge per unit length.



- Write down the differential contribution $d\vec{E}$ of an infinitesimal charge element dq located at position x on the rod to the electric field at $x = -d$. — *0.6 pts*
- What is the net electric field $\vec{E}$ at $x = -d$ due to the entire rod?. — *0.4 pts*

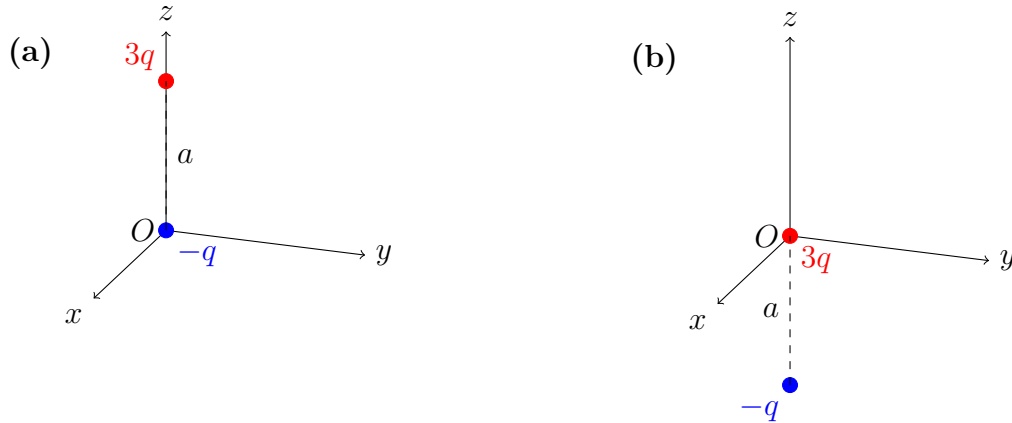
Given:

- $\int_0^{x_0} \frac{x dx}{(a+x)^2} = \left[\ln \left(\frac{a+x_0}{a} \right) + \frac{a}{a+x_0} - 1 \right]$, where $x_0 > 0$ and $a > 0$ are constants.

Solution:

Question 4

Two charges are placed in two different arrangements as shown below:



For each arrangement, determine

- (i) The monopole moment, — 0.2 pts
- (ii) The dipole moment, — 0.4 pts
- (iii) The approximate potential (in spherical coordinates) using these moments. — 0.4 pts

Given: The first two terms of multipole expansion of the potential is given by

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\vec{p} \cdot \hat{r}}{r^2} + \dots \right),$$

where Q is the monopole moment, $\vec{p}$ is the dipole moment. The spherical coordinates are given by (r, θ, ϕ) , where r is the distance from the origin, θ is the polar angle, and ϕ is the azimuthal angle.

Solution: